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Signals, Timing, Trust: Reflections from the Modular Peer Review Working Group

Final report from the Modular Peer Review Working Group

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Abstract

Between January and July 2026, the Continuous Science Foundation (CSF), PREreview, and more than 60 researchers, infrastructure builders, publishers, and community members came together as the Modular Peer Review Working Group. Together, we explored a simple but important question: *What might peer review look like if it evolved alongside the research process, rather than centering only on the final paper?*

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1. How we begun

As research becomes more open, many valuable research outputs—including study designs, datasets, code, figures, protocols, and models—are shared well before a manuscript is submitted for publication. Yet the way we evaluate research has remained largely organized around a single, static object: the paper. This disconnect raises important questions about how feedback could be provided earlier, how different research outputs might be evaluated on their own terms, and how trust in research could be built incrementally throughout the research lifecycle.

Recognizing this opportunity, CSF and PREreview convened the working group to imagine what a more modular approach to peer review could look like. Rather than designing a complete system from the outset, our goal was to generate practical, testable ideas that could help the community explore multiple possible futures for reviewing research outputs before —and beyond—the publication of a traditional manuscript.

We began by considering a wide range of research components, including datasets, methods, code, figures, notebooks, protocols, models, and other research artifacts. As the conversations progressed, the group focused on three concrete case studies—preregistrations, datasets, and figures—to collaboratively draft and refine review criteria that could be examined through both small- and large-group discussions.

The process was intentionally exploratory. Rather than seeking immediate consensus or a finished framework, we embraced experimentation, iteration, messiness, and productive disagreement as essential parts of the design process. By the end of six months, the working group produced not a final answer, but a flexible starting point: a shared foundation that can be built upon, tested, challenged, and adapted as the concept and implementation of modular peer review continue to evolve.

What follows is a session-by-session account of that process: what we did, what we produced, and what each session handed off to the next.

2. What we did, session by session

Session 1 — Naming the Objects

Question: *What could be reviewed?*

The group's first task was to identify everything in the research process that could be reviewed, evaluated, or receive feedback as its own independent object.

Participants from publishing, libraries, research, infrastructure, and open science communities across more than a dozen countries worked in small groups to generate as many ideas as possible. The exercise was intentionally expansive. There was no prioritization and no expectation of consensus—only an invitation to think broadly about where peer review might add value.

Participants identified a rich landscape of reviewable objects, including research design, ethics, protocols, datasets, software, computational workflows, analyses, figures, metadata, attribution, communication materials, and even peer reviews themselves.

Key outputs

- A categorized inventory of reviewable research objects.
- Early recognition that different objects require different kinds of expertise.
- Agreement that before designing review workflows, the group first needed to answer a more fundamental question: **what is the purpose of reviewing each object?**

That question became the focus of the second session.

Session 2 — Defining the Purpose

Question: *Why review it?*

With a broad landscape of reviewable objects established, the group shifted attention from *what* could be reviewed to *why* review happens in the first place.

Participants organized research objects into seven broad purposes for review:

- Knowledge exchange
- Improvement and iteration
- Validity and integrity
- Risk and responsibility
- Decision and resource allocation
- Recognition and credit
- Interoperability and standards

Working in breakout groups, participants explored which kinds of trust signals different reviews could generate. Looking across the discussions revealed a pattern: the same research objects—study design, datasets, code, and protocols—consistently appeared across multiple purposes. These emerged as particularly high-value opportunities for modular review.

The session concluded with a dot-voting exercise to identify where participants believed modular review could have the greatest impact. Preregistrations ranked highest, followed by datasets, code and computational workflows, funding proposals, and figures and visualizations.

Key outputs

- A framework describing seven purposes of peer review.
- Identification of five thematic working groups that would continue through the remainder of the project:
    - Preregistrations
    - Data
    - Code and notebooks
    - Funding proposals
    - Figures and visualizations

Having established why review matters, the next challenge became much more practical: **what would reviewers actually ask?**

Session 3 — Designing the Questions

Question: *How would someone actually review these objects?*

The third session translated abstract ideas into practical review criteria.

Participants began by reflecting on the most valuable feedback they had personally received as researchers. Across disciplines and career stages, similar themes emerged: useful reviews are specific, actionable, grounded in clear expectations, and respectful of authors' goals. Participants also identified common barriers, including incomplete context, unclear documentation, and the expectation that reviewers must be experts in every aspect of a project.

Three working groups—Preregistrations, Data, and Figures & Visualizations—then drafted object-specific review questions.

The Preregistration group distinguished preregistrations from registered reports and developed questions around research rationale, study design, statistical planning, and ethics. The Data group explored how review criteria change across raw, processed, and published datasets, emphasizing metadata, completeness, and reusability. The Figures group treated visualizations both as analytical products and communication tools, developing criteria spanning accuracy, accessibility, reproducibility, and clarity.

Key outputs

- Draft review questions for preregistrations, datasets, and figures.
- Conditions needed for those reviews to succeed.

Across all three groups, similar insights emerged. Effective review depends on clearly defining its purpose, providing sufficient context and metadata, and building on existing standards rather than creating entirely new ones.

These observations laid the groundwork for designing a broader review framework.

Session 4 — Connecting the Pieces

Question: *How do the pieces fit together?*

With review questions drafted, the working group turned to designing a conceptual framework describing how modular peer review might function.

The initial model followed a simple progression:

Object -> Function -> Criteria -> Outcome -> Signal -> Capture

This framework proposed that reviews begin with a specific research object, apply criteria appropriate to a defined purpose, generate trust signals, and ultimately capture those signals in ways that can travel with the object over time.

Participants then stress-tested the framework to improve it.

The Figures group observed that some trust signals—such as accessibility or licensing—could exist independently of an overall review outcome. The Data group argued that trust should be represented as a multidimensional profile rather than a single score. Meanwhile, the Preregistration group highlighted that many research objects are themselves composed of smaller reviewable objects, raising questions about nested review.

Key outputs

- A revised conceptual framework.
- The first proposal for multidimensional trust profiles, possibly visualized as radar charts.
- Recognition that review outcomes should move beyond binary judgments.

The session demonstrated the value of collaborative design, and actively reshaped the framework.

The next step was to test whether those ideas actually worked in practice.

Session 5 — Testing the Framework

Question: *Does it work?*

In response to feedback from Session 4, the working group simplified its framework to:

Object -> Function -> Criteria -> Signal -> Capture

Trust would no longer be represented as a single outcome, but as a multidimensional profile reflecting qualities such as robustness, reproducibility, clarity, usefulness, risk, care, compliance, and maturity.

Participants then conducted the group's first mock reviews using real research outputs: an open preregistration, a published figure together with its underlying data and code, and a publicly available dataset.

Working through structured review worksheets, each group identified the review's purpose, applied its draft criteria, evaluated resulting trust signals, and documented what information should be captured for future users.

Key outputs

- The first complete test of the modular review workflow.
- Evidence that review functions, criteria, and trust signals need explicit guidance rather than requiring reviewers to design them from scratch.
- New concepts emerging directly from practice, including reviewer confidence, feasibility, reuse, and clarity as distinct dimensions worth capturing.

While the exercise surfaced important challenges, it also demonstrated that modular review could be carried out in a practical and structured way.

Session 6 — Reflection and Looking Forward

Question: *What changed because of this work?*

The final session brought together six months of experimentation and reflection.

The group first revisited the full journey—from identifying reviewable objects to testing modular review with real research materials—before examining an external example that illustrated these ideas in practice.

PREreview shared lessons from its recent collaborative dataset review conducted in collaboration with the FORCE11 Club, where community participants used the recently developed dataset structured modular review workflow to review a dataset publicly available on Dryad. The pilot produced a published review and confirmed that many of the principles developed throughout the working group could be applied successfully in a real-world setting while also generating practical improvements to the review workflow itself.

Participants then reflected individually and collectively on three questions:

- What did we learn?
- What is emerging?
- What remains unfinished?

The resulting discussion moved beyond technical implementation toward broader questions about adoption, incentives, institutional support, reviewer recognition, interoperability, and communication.

Key outputs

- A community synthesis of six months of collaborative design.
- Identification of the major questions requiring continued experimentation.
- A shared commitment to continue developing and testing modular peer review beyond the lifetime of the working group.

Rather than ending with a finished framework, the project concluded with something more valuable: a community of practice committed to continuing the work together.

3. What We Learned

3.1. Modular peer review is feasible

Perhaps the most practical outcome of this collaboration was demonstrating that modular peer review can work.

Drawing on both the mock reviews developed throughout the working group and PREreview's live dataset review with the FORCE11 Club, participants showed that researchers can provide meaningful, constructive feedback on modular research outputs rather than only a traditional manuscript.

Just as importantly, testing review workflows in communities improves the workflows themselves. Each iteration produced better review prompts, clearer guidance for reviewers, and more actionable feedback for authors. Developing review processes openly also creates a virtuous cycle in which the workflows themselves continue to evolve through use and feedback.

3.2. Earlier feedback creates new opportunities

Throughout the six months, participants consistently returned to one idea: researchers benefit most from feedback when it arrives early enough to influence the work.

Rather than centering evaluation exclusively on the completed manuscript, participants envisioned review as an ongoing process alongside the research as it evolves. Earlier feedback can improve study design, strengthen datasets, clarify figures, identify potential issues before publication, and reduce costly backtracking later.

This enthusiasm extended beyond any single workflow. Participants consistently emphasized that peer review should become more continuous, constructive, and supportive of researchers as their work evolves.

3.3. An unexpected shift: from modularity to integrated review

Another outcome of the working group was one the organizers did not anticipate. We began by thinking of modularity primarily as reviewing different research objects using different review criteria.

The more we worked together, the clearer it became that the real opportunity might not lie in reviewing *more objects*, but in providing **the right feedback at the right time**. Modularity is but one way to enable integration of review into the research process.

Rather than imagining a separate review for every possible research output, participants increasingly described feedback integrated into key moments in the research lifecycle: planning a study, preparing a preregistration, collecting data, interpreting results, creating a single figure, communicating findings, and eventually sharing research narratives.

This shift reframed modular peer review as something less about fragmentation and more about continuity and timeliness. Individual review workflows remain important, but they become part of a larger progression of timely feedback that helps researchers throughout the life of a project.

3.4. Trust cannot be reduced to a single signal

Another idea surfaced repeatedly throughout the six months—and became particularly important as the group began designing "trust signals" for different kinds of reviews.

Initially, it seemed reasonable to ask what signals a review should generate. But participants quickly challenged the assumption that trust can be represented through a single score, badge, or universal set of criteria.

Trust means different things to different people and communities. A researcher deciding whether to reuse a dataset, a patient advocate evaluating whether research reflects community priorities, a funder assessing responsible stewardship, or a journal editor considering publication may all look for different evidence before deciding whether a research output is trustworthy or fit for purpose.

This realization shifted the conversation in two important ways.

First, participants moved away from thinking about trust as something a review could determine objectively. Instead, they began to see reviews as generating **contextual trust signals**—transparent pieces of information that help others make informed judgments for their own purposes. The idea of multidimensional trust profiles emerged from this discussion: rather than producing a single verdict, reviews might communicate multiple characteristics, such as clarity, reproducibility, accessibility, completeness, or methodological rigor, allowing different communities to interpret those signals according to their own needs.

Second, and perhaps more importantly, the group recognized that conversations about trust are also conversations about power.

For centuries, scientific institutions have concentrated authority to determine what counts as rigorous, credible, and trustworthy within relatively small groups of actors. Participants emphasized that modular peer review should not simply reproduce those same dynamics in a new technical system. Questions such as *Which trust signals matter? Who decides? Whose expertise counts? Whose perspectives are absent?* are not merely design decisions—they are questions of governance, representation, and equity.

Rather than prescribing a universal definition of trustworthiness, the working group came to see modular peer review as an opportunity to make evaluation more transparent, more contextual, and more open to multiple forms of expertise. This perspective aligns with the broader goal of building systems that distribute, rather than concentrate, the power to contribute to and evaluate scientific knowledge.

4. Questions We Left Open

Like any successful design process, this collaboration generated as many new questions as answers.

Among the questions participants identified for future exploration were:

- When is modular review most valuable?
- Which research objects should be prioritized?
- How can reviewers discover opportunities to contribute?
- What incentives encourage participation?
- How can trust signals be communicated without oversimplifying complex judgments?
- Where should modular reviews live?
- What metadata standards are needed to connect reviews with research objects?
- How should multidimensional trust profiles be represented and interpreted?
- What role can AI appropriately play in modular review?

Rather than viewing these as gaps, we see them as a shared research agenda for the broader peer review community.

5. What Comes Next

The working group's first six months established a foundation.

The immediate priority is to continue prototyping and testing review workflows for preregistrations, datasets, and figures in collaboration with researchers and partner organizations. Future work will focus not only on improving review criteria, but also on integrating modular review into existing infrastructure, metadata standards, and scholarly communication workflows.

Equally important will be developing incentives that encourage participation by researchers, reviewers, editors, funders, and institutions. Modular peer review should complement existing forms of peer review by providing timely, useful feedback earlier in the research process —not by creating additional burden or replacing established practices.

We also hope to expand this work through collaborative grant proposals, pilot projects, and partnerships across the scholarly communication ecosystem.

Many of the ideas presented here—including integrating timely peer-review, multidimensional trust profiles, and the recognition that trust itself is contextual rather than universal—emerged directly from participants' contributions and substantially changed the direction of the working group. Together we demonstrated the value of creating space for communities to think together, question assumptions, and reshape ideas through open collaboration.

Six months ago, we set out to imagine what it might mean to review research one piece at a time. We finished with a different realization: the greatest opportunity may be to provide the right feedback at the right moment throughout the life of a research project while recognizing that no single community should hold the authority to define what trustworthy science looks like for everyone.

That shift—from modularity as a collection of reviewable objects to peer review as a continuous process, contextual judgment, and shared responsibility emerged through conversation, experimentation, disagreement, and collaboration. This report captures not only what we built together, and is also an open invitation for others to help shape what comes next.